

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	QAS_L v2
People present	JEB, JJ, Maurice, Carsten Pilot
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	29, 30 Aug '22

Notes:

revisit 30 Aug '22

station moved 4 m East

maiken Revheim

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer CNR1	Y	Y	010343 4: 212891		Clean? ☒ Y / N
Ablation hose assembly	Y	N	50112 5002	same	No air in hose and bladder half full? Y / N
Iridium modem IMEI number	N		old reused		
Inclinometer/compass	N	old, yes	Attached to the radiometer? Y / N		
Satellite antenna	N	Y			
Wind sensor	Y	new	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	N	new hygro	Fan spins? ☒ Y / N Sounds, OK? ☒ Y / N clip		
HygroClip T/hum assembly	Y (new type)	clip yes	NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	Y	Y	Old sensor with new membrane? ☒ Y / N		
Sonic ranger on stake(s)	Y	Y	Old sensor with new membrane? ☒ Y / N NB: make sure not being buried next winter		
Thermistor string	Y (analog)	Y	shallowest @ -65 cm X -51 cm		
Solar panel	N	N	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N	N	At top of logger enclosure? ☒ Y / N		
Data logger	N	N	CR1000 or CR1000X?		
Multiplexer	N	N	Clicking during measurement cycle? ☒ Y / N		
Logger enclosure	N	N	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	N	N	New box? Y / N New desiccant bags? Y / N		
Stakes	Y	N re-used *	New one(s) drilled in? ☒ Y / N Length of new stake(s) 8 m? ☒ Y / N Bottom plugged to reduce melt-in? ☒ Y / N - is duct taped but old		
Tripod	N	N	New tripod? ☒ Y / N New sensor boom? ☒ Y / N		

re-used PTA,

* new stake arm, old was shorter carbon fiber, new is alu pipe, longer

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	+ 4 sec	+ 4 sec
Download data and/or change CF/SD Card (wait for green light) ✓	Y/N	Same CF Y/N
Name of logger program		Promice 245
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	- 1 °
Mast tilt across boom looking from radiometer (- if clockwise)	°	+ 1 °
Boom direction relative to north	° true/magn.	180 ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	3 ° W of S Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	Y/N
Temperature / humidity height (bottom of casing – surface)	cm	226 cm
Stake assembly standing?	Y/N / sort of	147 Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	147 cm
Free length of stake 2 (if 3: middle)	cm	carbon cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	1 cm
Height of ablation bladder fluid level above the ice surface	cm	156 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	- 9 * cm
Length of thermistor string on surface from surface marking	cm	- 51 cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

* PTA hole is lower
 AWS moved ~ 3 m to east to
 more level surface
 gathered CF card data, put it back in
 stake height 147 cm

Must-do at next visit:

new battery box? Zargos has
 "2010" on it!
 replace hygrocip
 " Rotronick, is partly
 broken

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	QAS_L v3 rewired
People present	JEB, JJ, Maurice pilot: carsten
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	29 August, 2022 + 30 Aug

Notes: Setup a complete new "v3 rewired"

site not finished day 1, returned 30 Aug
maiken present
tensioner was improvised

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	Y		212 886		Clean? ☒ Y / N
Ablation hose assembly Ablobender/normal PTA	Y		50054		No air in hose and bladder half full? Y / N
Iridium modem IMEI number	Y	New	3005340638	15540	
Inclinometer/compass Part of CNR4	Y	☒	Attached to the radiometer? Y / N		
Satellite antenna	Y	☒			
Wind sensor	Y	☒	Black box pointing toward the radiometer (i.e., along the boom)? ☒ Y / N		
Temperature / humidity assembly (lufft)	Y	☒	Fan spins? Y / N Sounds, OK? ☒ Y / N		
HygroClip T/hum assembly	N	X	NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Luft sensor (Y / N)		
Sonic ranger on AWS	Y	☒	Old sensor with new membrane? ☒ Y / N		
Sonic ranger on stake(s)	Y	Y	Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	Y (digital)		drilled in a little too deep?		
Solar panel	Y	☒	Clean? ☒ Y / N NB: is not effective when partially covered		
GPS antenna	Y		At top of logger enclosure? Y / N		
Data logger	Y		CR1000 or CR1000X?		
Multiplexer	N		Clicking during measurement cycle? Y / N		
Logger enclosure	Y		Replaced (including everything inside)? ☒ Y / N New desiccant bags? ☒ Y / N		
Battery box	Y		New box? ☒ Y / N New desiccant bags? ☒ Y / N		
Stakes	Y		New one(s) drilled in? ☒ Y / N Length of new stake(s) 8 m? ☒ Y / N Bottom plugged to reduce melt-in? ☒ Y / N		
Tripod	Y		New tripod? ☒ Y / N New sensor boom? ☒ Y / N		

* 50:05, meet start at sc

V2! free length stake 790 cm
old TC string was 5 m out

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	new AWS	
Download data and/or change CF/SD Card (wait for green light)	Y/N	Y/N
Name of logger program	promice.v3.rewired	1-1.CRX
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Boom direction relative to north	° true/magn.	180 ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	Y/N
Temperature / humidity height (bottom of casing – surface)	cm	277 cm
Stake assembly standing?	Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	— cm	162 cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	— cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	— cm
Length of thermistor string on surface from surface marking	cm	9 cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

▷ 8 m is too shallow for stake carbon because average ablation is 6.5 m stake installed 6.38 m and is only

Must-do at next visit:

10 m carbons

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	QAS-M v2
People present	JB, JJ, Maiken, Maurice
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	30 Aug 2022 → reposition ~2km north

Notes: Relocate v2 and Snowfox

Take down GNSS station

maurice landed with us, returned with stuff to UAK to offload and pick up BBC radio
 Program not running? no public variables, no data?

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer CNR1	Y	212889	080023 ✓		Clean? Y / N
Ablation hose assembly	N	50205			No air in hose and bladder half full? Y / N
Iridium modem IMEI number	N	N	modem switch from old logger		
Inclinometer/compass	N	not present	Attached to the radiometer? Y / N		
Satellite antenna	N	✓			
Wind sensor	N	✓	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	N		Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	Y (old type)	changed	NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? old NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	Y	X	Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	Y	X	Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	Y (analog)	Y			
Solar panel	N	Y	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N	X	At top of logger enclosure? Y / N		
Data logger	Y	Y	CR1000 or CR1000X?		
Multiplexer	N	X	Clicking during measurement cycle? Y / N		
Logger enclosure	N	New	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	N	X	New box? Y / N New desiccant bags? Y / N		
Stakes	Y	X	New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N 10 m Bottom plugged to reduce melt-in? Y / N sort of		
Tripod	Y	Y	New tripod? Y / N New sensor boom? Y / N		

Problems:
 1. SR50 cable unplugged on arrival
 2. boom arms bent down ~10°

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3. Hibo rain tilting ~40°

4. pin on TC strings pulled
 plug on logger box

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	+50 sec	4 s
Download data and/or change CF/SD Card (wait for green light)	☒ Y / N	☒ Y / N
Name of logger program	Prom:CC2015E	
Screen dump of values in fast scan mode	☒ Y / N	☒ Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / N	☒ Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	+ 4 °
Mast tilt across boom looking from radiometer (- if clockwise)	°	+ 7 °
Boom direction relative to north	150 ° true/magn.	150 ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	☒ Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	NA Y / N
Temperature / humidity height (bottom of casing – surface)	cm	237 cm
Stake assembly standing?	Y / N / sort of	☒ Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	583 $\pm$ 25 cm	415 $\pm$ 25 cm
Free length of stake 2 (if 3: middle)	cm	/ cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	/ cm
Height of ablation bladder fluid level above the ice surface	cm	154.5 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	+ 30 * * cm
Length of thermistor string on surface from surface marking	cm	+ 4 cm
Apparent damage to sensors? If yes, which?	Y / N	X * X ☒ Y / N

Notes:

* Stake leaning 20°, surface undulations
ice height 19.02 when station op $\triangleq$ arrival
stake height 00 not working
* * hole is higher than under the tug
* * * Rotronics bent

Must-do at next visit:

- inclinometer
- replace rotronic that is bent downward 150

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	QAS_U v3 alpha
People present	
Purpose	Maintenance / installation / <u>take-down</u> / quick visit / emergency
Date	

Notes: Take down station

boom arms bent down, radiometer -30.2° to south
mast lean -15.0° to south measured

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N		number invisible		Clean? Y/N <u>bent down</u>
Ablation hose assembly	N	<u>not installed</u>			No air in hose and bladder half full? Y/N
Iridium modem IMEI number	N	5154	530		
Inclinometer/compass	N		Attached to the radiometer? Y/N		
Satellite antenna	N	Y			
Wind sensor	N	✓	Black box pointing toward the radiometer (i.e., along the boom)? Y/N		
Temperature / humidity assembly	N		Fan spins? Y/N Sounds, OK? Y/N <u>1u fff</u>		
HygroClip T/hum assembly	N	✓	NB: Rotronics (Y/N) could have a 40 °C offset. New or old type? NB: Luft sensor (Y/N)		
Sonic ranger on AWS	N		Old sensor with new membrane? Y/N		
Sonic ranger on stake(s)	N		Old sensor with new membrane? Y/N NB: make sure not being buried next winter		
Thermistor string	N	<u>none</u>			
Solar panel	N	✓	Clean? Y/N NB: is not effective when partially covered		
GPS antenna	N	✓	At top of logger enclosure? Y/N <u>duct taped</u>		
Data logger	N	✓	CR1000 or CR1000X?		
Multiplexer	N	✓	Clicking during measurement cycle? Y/N		
Logger enclosure	N	✓	Replaced (including everything inside)? Y/N New desiccant bags? Y/N <u>no desiccant</u>		
Battery box	N	✓	New box? Y/N New desiccant bags? Y/N <u>larger one</u>		
Stakes	N	✓	New one(s) drilled in? Y/N Length of new stake(s) 8 m? Y/N Bottom plugged to reduce melt-in? Y/N		
Tripod	N	✓	New tripod? Y/N New sensor boom? Y/N		

QAS U V3
alpha beta

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	- 204UTC	
Download data and/or change CF/SD Card (wait for green light)	Y/N	Y/N
Name of logger program	Promice V3 D	
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	+ 15.0 °	°
Mast tilt across boom looking from radiometer (- if clockwise)	+ 10.0 °	°
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	Y/N
Temperature / humidity height (bottom of casing – surface)	cm	cm
Stake assembly standing?	Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	458.5* cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

* 7 to 18 cm superimposed ice,
patchy so sonic^{SRSO} will integrate
radiom 10° W of South

Must-do at next visit:

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	QAS_U v2
People present	JEB, JJ, Maurice, Maiken
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	31 Aug 2022

Notes: Replace battery on snowfox

boom arms bent down
mast lean -6°, radiometer -15.6° to south
hydraulic bag hanging out, door had opened

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	N			Clean? ☒ Y / N
Ablation hose assembly	N				No air in hose and bladder half full? Y / N
Iridium modem IMEI number	N				
Inclinometer/compass	N		Attached to the radiometer? Y / N		
Satellite antenna	N				
Wind sensor	N		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	N		Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	Y (new type)		NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	Y		Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	Y		Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	N				
Solar panel	N		Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N		At top of logger enclosure? Y / N		
Data logger	N		CR1000 or CR1000X?		
Multiplexer	N		Clicking during measurement cycle? ☒ Y / N		
Logger enclosure	N		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	N		New box? Y / N New desiccant bags? Y / N		
Stakes	Y		New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	N		New tripod? Y / N New sensor boom? Y / N		

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	+35 sec	
Download data and/or change CF/SD Card (wait for green light)	Y/N	Y/N
Name of logger program	Promice 2015 E	
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	+6 °	°
Mast tilt across boom looking from radiometer (- if clockwise)	0 °	°
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	Y/N
Temperature / humidity height (bottom of casing – surface)	cm	cm
Stake assembly standing?	Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	463* cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	114 cm	151 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	-8 cm
Length of thermistor string on surface from surface marking	cm	475 cm
Apparent damage to sensors? If yes, which?	*XRo tronic is in need Y/N	ish Y/N

Notes: x auf eis, 7 to 18 cm thick,
aka "superimposed ice", see photos
radiom pointing 9° w of true S

Must-do at next visit:

replace SR50 on boom is
SR50 address issue to be aware of?
** abl hose box is taped shut,
not really a problem

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	QAS_U v3 rewired
People present	
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	

Notes: Setup a complete new "v3 rewired"

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	Y		213257		Clean? Y / N
Ablation hose assembly	Y				No air in hose and bladder half full? Y / N
Iridium modem IMEI number	Y				
Inclinometer/compass	Y		Attached to the radiometer? Y / N		
Satellite antenna	Y				
Wind sensor	Y		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly (lufft)	Y		Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	N		NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	Y		Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	Y		Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	Y (digital)				
Solar panel	Y		Clean? Y / N NB: is not effective when partially covered		
GPS antenna	Y		At top of logger enclosure? Y / N		
Data logger	Y		CR1000 or CR1000X?		
Multiplexer	Y		Clicking during measurement cycle? Y / N		
Logger enclosure	Y		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y		New box? Y / N New desiccant bags? Y / N		
Stakes	Y		New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	Y		New tripod? Y / N New sensor boom? Y / N		

QASU
V3rewired

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		- 00:04
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program		PRATICE V3rewired
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	-2,5 °
Mast tilt across boom looking from radiometer (- if clockwise)	°	-3,7 °
Boom direction relative to north	° true/magn.	196 ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	Y / N
Temperature / humidity height (bottom of casing – surface)	cm	282 cm
Stake assembly standing?	Y / N / sort of	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	415 cm
Free length of stake 2 (if 3: middle)	cm	17 cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	170 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	-10 cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y / N	Y / N

Notes:

⊕ The boom is levelled $\sim \pm 0,5^\circ$
~~no re-drilling of stake~~
 new stake drilled closer to station
 from 1st thermistor

Must-do at next visit:

Apr 2 1 Sept 2022

- 8m 64 cm btw antennae
- solar is 73 cm above surface, bottom of
- JJ, JEB, fila
- photos taken

Q5, patchy snow

other stake, taller left 119 cm

2021? new red + blk flag on 4m

or was it the 6m with not much

flag that fell

← kae ala mast

5x pipes daily in contact with

fell from that one?

Total DKK 3,040.00

28-08-22	Hotel Narsarsuaq, Double room, Shared facilities	2	Pax
29-08-22	Hotel Narsarsuaq, Double room, Shared facilities	2	Pax
28-08-22	Wool Blanket	1	pcs.
22-08-22	Camping Gas- 4x430	4	pcs.

Date	Description	Quantity	Unit of Measure	Amount
28. August 2022	8618			
11. September 2022				
28. August 2022				
Document Date				
Due Date				
Posting Date				
Invoice No.				
Bill-to Customer No.	979			
Sell-to Contact				
Phone No.				
E-Mail				
Home Page				
VAT Reg. No.				
Salesperson				
Bank				
Account No.				
IBAN				
SWIFT Code				

installed new 6m - 0.22m high + red black flag re used, new height +

east of currently 3

west

east

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www.blueice.gl
12001363
Sten Svendsen
GrønlandsBANKEN
6471 0002013832
GL5864710002013832
GRENGLGX

Blue Ice Explorer
P.O. Box 58
Grønland
3923 Narsarsuaq

Sales - Invoice COPY

older stake, leaning to south

red/blk flag

8.95 high

with 6m

junction - 2

carbon

Q4



QAS-3

30 Aug

JEB-55
marker

propeller gone, Luft
wind baffles partly fallen off.

logger data only since 22 June '22 -
same issue last time

4.6 cm fluid level @ centre of bucket